

BIOLOGY — GRADE 10 | PLANT NUTRITION

Quick Check

Lesson 2: Leaf Structure: The Architecture of a Photosynthetic Machine

10 multiple-choice questions • choose A, B, C or D

Sub-Strand 2.1: Plant Nutrition | Lesson 2

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You are looking at a cross-section of a mango leaf. On which layer would you find most of the stomata, with a pair of guard cells around each one?

- A** The palisade mesophyll
- B** The waxy cuticle on the upper surface
- C** The xylem in the vascular bundle
- D** The lower epidermis

In a leaf cross-section, which layer sits just below the upper epidermis and is made of tightly packed, column-shaped cells full of chloroplasts?

- A** Lower epidermis
- B** Spongy mesophyll
- C** Palisade mesophyll
- D** Phloem

The spongy mesophyll has many air spaces between its cells. What is the main advantage of these air spaces?

- A** They carry sugars away from the leaf to the stem
- B** They absorb sunlight that passes through the epidermis
- C** They store extra water for the leaf to use at midday
- D** They let carbon dioxide diffuse easily to the cells inside the leaf

A vascular bundle in a leaf contains xylem and phloem. Which statement describes their jobs correctly?

- A** Xylem brings water and minerals to the leaf and phloem carries sugars away
- B** Xylem carries sugars out of the leaf and phloem brings water into the leaf
- C** Xylem and phloem both carry only carbon dioxide to the palisade cells
- D** Xylem lets in light and phloem controls the opening of stomata

The cuticle and the epidermis of a leaf are transparent (clear). Why is this useful to the plant?

- A** It allows water vapour to escape freely from the whole leaf surface
- B** It lets air spaces fill with sunlight at night
- C** It helps the xylem to see where water is needed
- D** It lets light pass through to the chloroplasts in the cells below

What is the job of the two guard cells around a stoma?

- A** To protect the leaf by making a thick waxy layer
- B** To carry water from the roots to the leaf
- C** To open and close the stoma, controlling gas exchange and water vapour loss
- D** To make sugars using sunlight

At midday in bright sun, a kale plant's stomata are wide open. The soil is still moist, yet the leaves wilt. Which explanation fits the leaf structure best?

- A** The guard cells close completely and trap heat inside the leaf
- B** Water vapour leaves through the open stomata faster than the xylem can replace it, so the cells lose turgor
- C** The roots have permanently died, so no water can enter the plant
- D** Too much water enters the leaf cells and they burst

A kale plant is moved under the shade of a mango tree. Wilting stops, but over several weeks the leaves turn pale and the plant grows slowly. Which explanation uses leaf structure correctly?

- A** The shade blocks the xylem, so no water can reach the leaf
- B** The shade makes the guard cells lose their pores
- C** Less light reaches the palisade cells, so the leaf makes less food and the plant grows slowly
- D** The shade makes the cuticle so thick that air cannot pass through

A student is revising her model of a kale leaf in a cross-section drawing. Which set of arrows is correct?

- A** Light enters through the stoma, water moves down the xylem, and water vapour leaves through the vascular bundle
- B** Light enters from above through the epidermis, water moves up the xylem, and water vapour leaves through a stoma
- C** Light enters from below, water moves down the phloem, and water vapour enters through the cuticle
- D** Light enters from above, sugars move up the xylem, and water vapour leaves through the palisade cells

After observing a leaf cross-section, which Driving Question Board question could students already partly answer using today's evidence?

- A** Why do sukuma wiki leaves wilt at midday even when the soil is moist?
- B** What exactly makes the guard cells open the stoma?
- C** Why are leaves green?
- D** What is the full chemical equation for photosynthesis?

All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).